Book 1 — Lecture 1 — Exercise 1

Closed vs Open Systems

Prompt. Since the notion is so important to theoretical physics, think about what a closed system is and speculate on whether closed systems can actually exist. What assumptions are implicit in establishing a closed system? What is an open system?

Closed System (working definition)

In theoretical physics, a *closed* system is one you describe with a *complete* set of dynamical variables and *deterministic* (or, in quantum theory, unitary) laws that involve *only* those variables. Nothing outside the model is allowed to influence the state except through quantities you have already included.[1]

Implicit Assumptions

Calling a system "closed" usually includes several idealizations:

- **Exhaustive state:** all relevant degrees of freedom are modeled.[2]
- **Known laws:** dynamics is assumed exact for that state.
- **Isolation:** outside influences are negligible or explicitly modeled.
- **Boundary in time and space:** closure is only approximate over a chosen window.

Can Closed Systems Actually Exist?

Operationally, we can approximate closure very well. Fundamentally, perfect closure is doubtful: long-range gravity, environmental coupling, and quantum entanglement are hard to eliminate completely. In practice, "closed" is a controlled approximation whose quality depends on the required precision.

Open System

An *open* system exchanges energy, momentum, particles, or information with an environment that is not fully tracked. Its effective dynamics then includes dissipation, noise, driving, or

[1] A *dynamical variable* is a quantity whose value changes with time and helps specify the system state. A law is *deterministic* when the current state fixes a unique next state.

[2] A *degree of freedom* is an independent parameter needed to describe a system configuration.

explicit coupling terms. Often the combined "system + environment" is treated as closed, while the subsystem of interest is open.[3]

Result. Classical mechanics often starts with closed deterministic models. When that is insufficient, one either enlarges the modeled system or uses open-system descriptions.

Example. A pendulum in vacuum over short times can be modeled as nearly closed; adding air drag and thermal fluctuations turns it into an open-system description.

[3] A *subsystem* is a selected part of a larger physical system. *Dissipation* means irreversible energy transfer (often to heat).